Network-Based Asset Clustering vs GICS Sector Classifications: Evidence from the S&P 500 Across Economic Regimes

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ABSTRACT

Portfolio diversification through GICS sectors assumes that intra-sector stocks share risk profiles while inter-sector stocks provide genuine diversification, yet the simultaneous collapse of all eleven GICS sectors during the 2008 Global Financial Crisis and the 2020 COVID-19 crash suggests that this assumption fails precisely when diversification is most needed. This study constructs Planar Maximally Filtered Graphs from GARCH-standardised correlation matrices of S&P 500 stocks across ten economic regimes from 1998 to 2026, partitions each network using the Louvain optimisation method, and compares the resulting stock-level partitions to GICS sector classifications using Mutual Information. Network partitions share substantial information with GICS sectors; however, partition structures are highly unstable across regimes, suggesting that while GICS classifications are meaningful summaries of the underlying correlation structure during stable market regimes, the correlation structure is fundamentally regime-dependent, with direct implications for risk management.

I. INTRODUCTION

Portfolio diversification is conventionally achieved through sector-based allocation using the Global Industry Classification Standard (GICS), a framework developed jointly by MSCI and Standard & Poor's that categorises every listed equity into one of eleven sectors based on its primary business activity. Risk managers, index providers, and asset allocators rely on GICS under two assumptions: (i) companies within the same sector share similar risk exposures, so concentration within a sector is a source of idiosyncratic risk, and (ii) companies across different sectors provide genuine diversification, so capital spread across sectors is protected against sector-specific shocks.

However, empirical evidence from financial crises calls these assumptions into question. During the 2008 Global Financial Crisis, all eleven GICS sectors experienced severe drawdowns, with the S&P 500 declining approximately 57% from peak to trough. During the 2020 COVID-19 crash, the index fell 34% in 23 trading days. These episodes suggest that, under market stress, the correlation structure of equities may reorganise in ways that render sector boundaries irrelevant. For portfolio managers, this phenomenon is consequential: risk models that assume stable sector correlations will systematically underestimate portfolio risk during precisely the episodes where accurate risk estimation matters most.

Network-theoretic approaches offer an alternative, data-driven view of asset co-movement. Mantegna [1] introduced the Minimum Spanning Tree (MST) as a tool for filtering pairwise correlation matrices into a tractable network structure. Tumminello et al. [2] extended this with the Planar Maximally Filtered Graph (PMFG), which retains the MST as a subgraph while preserving substantially more edges through a planarity constraint. Community detection algorithms such as the Louvain method [3] identify data-driven clusters within these networks based on observed correlation behaviour rather than predetermined industry labels.

This report extends the work conducted for the group presentation, which analysed dynamic correlations between eleven S&P 500 sector ETFs over the period 1998–2026. The group project employed Dynamic Conditional Correlation (DCC)-GARCH models (a multivariate extension of GARCH that estimates time-varying pairwise correlations [4]) alongside MST network construction at the sector level. Key findings from the group project included: (i) DCC-based correlations are substantially more stable than rolling Pearson correlations during high-volatility episodes; (ii) the Industrial sector consistently exhibits the highest betweenness centrality in the sector-level MST, acting as the primary intermediary through which shocks propagate between sectors; and (iii) lead-lag relationships between sectors, validated by Granger causality tests, are strongly regime-dependent. The present study moves from the sector-ETF level to the individual stock level, analysing approximately 500 stocks per regime, to address a more fundamental question: does the GICS classification itself accurately reflect the underlying stock-level co-movement structure of equities, and does this relationship change across economic regimes?

II. DATA

A. Data Source and Universe

Daily close prices for all current and historical S&P 500 constituents are obtained from Refinitiv via the LSEG Data API, covering 2 January 1998 to 1 April 2026. Refinitiv is selected because it retains price data for delisted securities, ensuring that analysis is free from the survivorship bias that would arise from relying only on currently listed securities. The full dataset covers 1,096 unique instruments with price data, including 503 current index members (Alphabet, Fox and News Corp each list two share classes of stock) and 593 historical constituents that were removed during the sample period.

GICS sector classifications are obtained directly from Refinitiv for 692 of 1,096 stocks. The remaining 404 stocks (predominantly long-delisted companies) are imputed via a cascading TRBC (The Refinitiv Business Classification)-to-GICS mapping built from the 692 stocks with both classifications. For each TRBC level, the modal GICS sector is identified and applied in order of increasing granularity reduction (Activity $\rightarrow$ Industry $\rightarrow$ Industry Group; mean intra-Activity agreement = 98.4%). A further 8 stocks with ambiguous TRBC Activities ("Investment Holding Companies" or "Shell Companies") are manually classified from their historical operating businesses.

Index membership is tracked using a joiners and leavers dataset exported directly from Refinitiv Workspace, comprising 2,147 events since 30 December 1994. Membership intervals for each RIC are constructed from the event log. Where leave-and-rejoin events occur within 20 calendar days, these are merged as they typically reflect corporate restructuring (ticker changes, share class reclassification) rather than genuine exits.

For each regime, a stock is included in the correlation matrix if it satisfies two conditions: (i) it was an S&P 500 member at the start and/or end of the regime; and (ii) it has complete price data (at most 5% missing observations after forward-filling gaps of up to three trading days) over the full regime period. Stocks that both entered and exited the S&P 500 within the same regime are excluded as their brief membership is insufficient for meaningful correlation estimation.

TABLE I. Regime definitions with period boundaries, type classification (based on NBER business-cycle dates and major market events), number of trading days, and number of stocks included for analysis.

Regime	**Period**	**Type**	**Days**	**N**
Dot-com Build-up	1998-01-01 to 2000-03-09	Expansion	551	483
Dot-com Bust	2000-03-10 to 2002-10-09	Crisis	648	515
Expansion / Credit Build-up	2002-10-10 to 2006-12-31	Expansion	1,064	506
Housing Bubble Peak	2007-01-01 to 2007-12-31	Pre-crisis	251	503
Global Financial Crisis	2008-01-01 to 2009-06-30	Crisis	377	516
Post-GFC Expansion	2009-07-01 to 2019-12-31	Expansion	2,644	516
COVID Crisis	2020-01-01 to 2020-06-30	Crisis	125	508
Post-COVID Recovery	2020-07-01 to 2021-12-31	Expansion	380	518
Inflation / Tightening Shock	2022-01-01 to 2023-12-31	Crisis	501	516
Post-Tightening / Recent	2024-01-01 to present	Expansion	564	523

B. Statistical Properties of Returns

Returns exhibit the standard stylised facts of financial time series [5]. Across 522 long-lived stocks (those with data coverage exceeding 80% of the 1998–2026 period), median excess kurtosis is 13.27 and median skewness is -0.30, motivating the use of Student-t innovations in the volatility model. Volatility clustering is visible in the rolling 252-day average pairwise correlation (Figure 1), which is strongly regime-dependent: the mean across crisis regimes (0.42) is 1.7 times the mean across expansion regimes (0.25).

III. METHODOLOGY

A. Volatility Modelling: Univariate GARCH(1,1)-t

Volatility clustering is established as a central stylised fact of financial returns [5], and GARCH provides the standard parametric approach for capturing this phenomenon. We fit a univariate GARCH(1,1) model with Student-t innovations [6] to each stock's return series

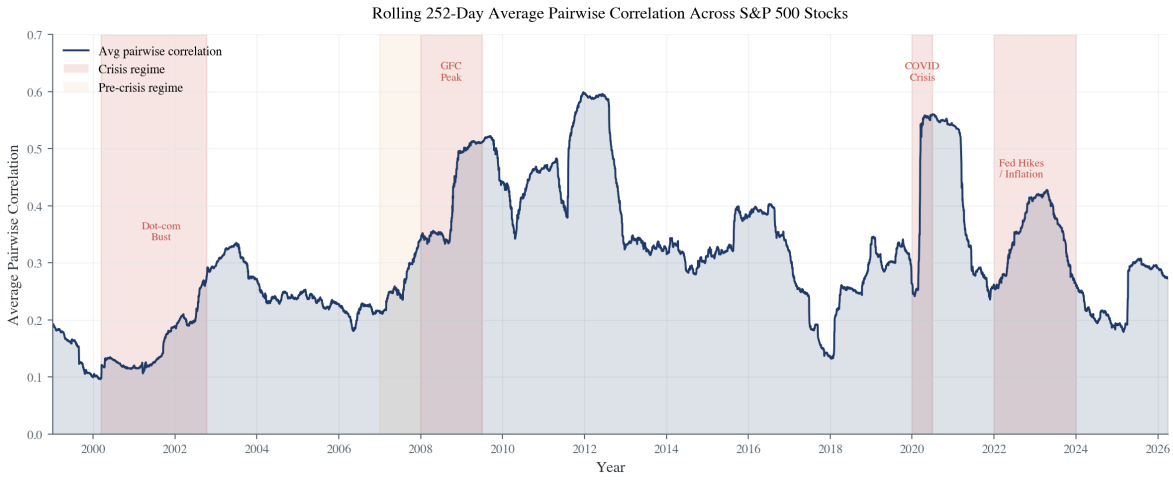


FIG. 1. Rolling 252-day average pairwise correlation across a 100-stock random sample of long-lived S&P 500 stocks. Crisis regimes are shaded in red, the pre-crisis Housing Bubble Peak in amber. Correlation reaches its highest values during the GFC (> 0.6) and remains elevated through the 2010–2012 European sovereign-debt period.

within each regime. The model specification is:

$$r_{it} = \mu_i + \epsilon_{it} \tag{1}$$

$$\epsilon_{it} = \sigma_{it}\, z_{it} \tag{2}$$

$$\sigma_{it}^2 = \omega_i + \alpha_i \epsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2 \tag{3}$$

$$z_{it} \sim t(\nu_i) \tag{4}$$

where r_{it} is the log return of stock i at time t, μ_i is the conditional mean, σ_{it}^2 is the conditional variance, and z_{it} are standardised Student-t innovations with ν_i degrees of freedom [6, 7].

The standardised residuals $z_{it} = \epsilon_{it}/\sigma_{it}$ form the basis for all subsequent correlation analysis. Dividing by the conditional volatility removes the time-varying volatility component, so that the resulting correlations reflect genuine co-movement rather than shared volatility dynamics. Empirically, raw Pearson correlation overstates the mean pairwise correlation by 0.022 on average across regimes, with the largest distortion (+0.084) during the COVID Crisis.

B. Information Filtering: MST and PMFG

Pearson correlation ρ_{ij} takes values in $[-1, 1]$ but does not satisfy the triangle inequality required for a metric. To construct a proper metric space, correlations are transformed into Mantegna distances $d_{ij} = \sqrt{2(1 - \rho_{ij})}$ [1], which satisfy non-negativity, symmetry, and the triangle inequality. The full stock-level correlation matrix for $N \approx 500$ stocks contains $N(N-1)/2 \approx 125{,}000$ pairwise values, most of which represent noise; information filtering extracts the structurally significant edges.

The Minimum Spanning Tree (MST) is a connected, acyclic subgraph spanning all N nodes with minimum total Mantegna distance, constructed using Kruskal's algorithm in $O(E \log E)$ time where $E = N(N-1)/2$ is the number of candidate edges. The MST retains exactly $N - 1$ edges and captures the dominant correlation skeleton, but its extreme sparsity makes it unsuitable for partition detection: the tree structure forces partitions to be subtrees, with no internal loops or cliques.

The Planar Maximally Filtered Graph (PMFG) [2] addresses this limitation. The PMFG is the maximal planar subgraph containing the MST as a subgraph: edges are added in descending order of correlation subject to the constraint that the resulting graph remains planar (embeddable on a plane without edge crossings). By Euler's formula for planar graphs, the PMFG retains exactly $3(N - 2)$ edges, approximately three times the MST. The PMFG's richer edge set preserves the loop and clique structures that are essential for meaningful partition detection.

The PMFG is chosen over simple correlation-threshold networks (where all edges with $|\rho_{ij}| > c$ are retained) because the planarity constraint provides a principled, parameter-free filtering criterion. Threshold networks are sensitive to the arbitrary cutoff c: too low retains noise, too high produces disconnected components. The PMFG is chosen over MSTs for partition detection for the reasons outlined above; MSTs are computed and reported as a comparison baseline.

C. Network Partitioning

Having constructed a PMFG for each regime, we partition the network into groups of densely-connected nodes. This partitioning provides a data-driven grouping of stocks that

can be compared to the GICS classification.

We use the Louvain algorithm [3] to identify partitions that maximise modularity:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{5}$$

where A_{ij} is the adjacency matrix, k_i is the degree of node i, m is the total number of edges, and $\delta(c_i, c_j)$ equals 1 if nodes i and j share a partition and 0 otherwise. The Louvain algorithm and modularity as a partition quality metric [8] are standard tools for this task. The algorithm is run 10 times with different random seeds and the partition achieving the highest modularity is retained.

D. Comparison Framework

The core quantitative comparison between network partitions and GICS sectors uses Mutual Information (MI), an information-theoretic measure of dependence between two random variables. Treating each partition as a random variable over the stock universe, MI is:

$$I(U;V) = \sum_{u} \sum_{v} p(u,v) \, \log \frac{p(u,v)}{p(u)\, p(v)} \tag{6}$$

where U is the network partition and V is the GICS partition. $I(U;V) = 0$ if and only if the two partitions are independent; larger values indicate more shared information. MI is unbounded above, with any-sample maximum bounded by $\min(H(U), H(V))$, where H denotes entropy.

Statistical significance and confidence intervals. We assess significance by shuffling the GICS labels 10,000 times per regime and recomputing MI; the p-value is the fraction of null MIs equalling or exceeding the observed value. Empirically, the observed MI for the Post-GFC Expansion (1.53) sits far above the null distribution mean (0.12), yielding $p < 0.0001$; all ten regimes yield $p < 0.0001$. For each observed MI, a 95% confidence interval is additionally computed by resampling stocks with replacement 1,000 times.

E. Lead-Lag Analysis and Granger Causality

To assess temporal dependencies between the detected communities (i.e., whether the returns of one community contain predictive information about another community's future

returns) we extend the cross-sectional correlation analysis with a lead-lag framework based on cross-correlation at lags and the Granger causality test [9].

Community returns. For each regime, each detected community c yields a community return series $r_t^{(c)}$ computed as the equal-weighted mean of the GARCH-standardised residuals of its member stocks. Only communities with at least three members are retained.

Cross-correlation at lag k. For each ordered pair of communities (X, Y), the lagged cross-correlation $\rho_{XY}(k) = \mathrm{Cov}(X_t, Y_{t+k})/(\sigma_X \sigma_Y)$ is evaluated for lags $k = 1, 2, \ldots, 21$ trading days (approximately one month). For each pair, the lag k^* maximising $|\rho_{XY}(k)|$ is retained as a candidate lead-lag relationship.

Granger causality test. Given the candidate lag k^*, we formally test whether community X Granger-causes community Y using a standard F-test on nested vector autoregressions at lag k^* [9], rejecting at $\alpha = 0.05$. A directed edge $X \to Y$ is retained in the leadership network when the null is rejected.

Leadership network. For each regime, the directed edges form a leadership graph. The net leadership of community c is defined as:

$$\mathrm{NetLeadership}(c) = |\{Y : c \text{ Granger-causes } Y\}| - |\{X : X \text{ Granger-causes } c\}| \qquad (7)$$

Positive values identify communities whose returns contain predictive information about others; negative values identify communities that follow others. Applying this at the community level rather than the stock level ensures that the number of pairs is tractable (~ 150 per regime) and the results are interpretable (each community is labelled by its dominant GICS sector in the results).

IV. RESULTS

A. Network-GICS Alignment

Table II reports the Mutual Information (MI) between PMFG partitions and GICS sector classifications for each regime. The mean MI is 1.20, with all ten values statistically significant at $p < 0.0001$ under the bootstrap null-distribution test. For reference, a random partition of similar sizes would yield MI approximately 0.1, placing our observed values an order of magnitude above chance. The Louvain algorithm detects on average 12.2 partitions, compared to 11 GICS sectors, suggesting the data support a slightly finer-grained grouping

than the GICS framework provides. Mean modularity is 0.82, indicating strong partition quality.

Two regimes stand out. The Post-GFC Expansion achieves the highest alignment (MI = 1.53) despite being the longest regime (10.5 years); this reflects the extended period of relatively stable factor structure following 2009. The Housing Bubble Peak shows the weakest alignment (MI = 0.80) despite being a pre-crisis rather than crisis period. This reflects the specific dynamics of late-cycle bubble behaviour: financial innovation, the rapid growth of mortgage-backed securities and structured credit products, and the cross-sector reach of housing exposure (spanning Financials, Real Estate, Consumer Discretionary, and Materials) produced unusual co-movements that blurred traditional industry boundaries.

TABLE II. Community detection results by regime. MI is the primary partition-comparison metric; all values are statistically significant at $p < 0.0001$ under bootstrap label-shuffling. N is the number of stocks in the regime, Parts the number of Louvain partitions, and Q the PMFG modularity.

Regime	**N**	**Parts**	**Q**	**MI**	**95% CI (MI)**
Dot-com Build-up	483	10	0.79	1.00	[0.95, 1.13]
Dot-com Bust	515	12	0.84	1.20	[1.16, 1.33]
Expansion / Credit Build-up	506	11	0.81	1.16	[1.11, 1.30]
Housing Bubble Peak	503	10	0.79	0.80	[0.77, 0.96]
Global Financial Crisis	516	13	0.80	1.35	[1.30, 1.48]
Post-GFC Expansion	516	12	0.83	1.53	[1.47, 1.63]
COVID Crisis	508	13	0.81	1.09	[1.04, 1.23]
Post-COVID Recovery	518	15	0.85	1.28	[1.23, 1.42]
Inflation / Tightening	516	13	0.85	1.36	[1.31, 1.48]
Post-Tightening / Recent	523	13	0.84	1.23	[1.19, 1.36]

B. Cohesion Comparison

The intra/inter-partition correlation ratio is computed for both the network partition and the GICS partition. Network partitions achieve a higher ratio in 4 of 10 regimes, with a mean network ratio of 1.46 compared to 1.44 for GICS. In absolute terms, the two partitions

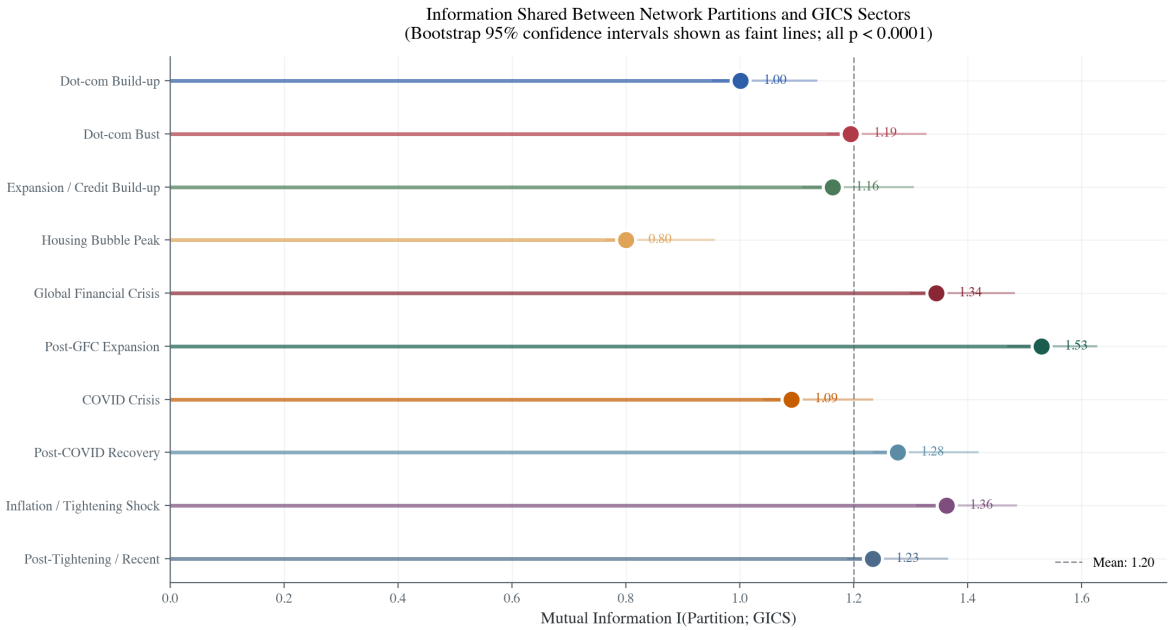


FIG. 2. Mutual Information (MI) between PMFG-detected partitions and GICS sectors across the ten regimes, with bootstrap 95% confidence intervals. The Post-GFC Expansion shows the strongest alignment (MI = 1.53), the Housing Bubble Peak the weakest (MI = 0.80); all values are statistically significant at $p < 0.0001$.

produce very similar cohesion ratios (consistent with their mean MI of approximately 1.20), identifying similar structures, with network partitions providing marginal improvements in more recent regimes, particularly the Post-COVID Recovery (network − GICS = +0.255) and Post-Tightening / Recent periods.

C. Partition Instability Across Regimes

The most striking finding is the low cross-regime stability. Pairwise MI between PMFG partitions of different regimes averages 1.12 (off-diagonal mean of the stability matrix, computed on the set of stocks common to both regimes). Adjacent regimes, which share market conditions most closely, have mean MI of 1.17, only marginally higher than the overall average. Between distant regimes such as the Dot-com Build-up and the Post-Tightening Recent period, MI falls to approximately 0.80.

This instability exists despite strong within-regime modularity (mean 0.82). The implication is that while partitions are well-defined within any given regime, the identity of

partitions (which stocks group with which) shifts fundamentally between regimes (Figure 3). Figure 4 illustrates this reorganisation directly, showing how communities merge, split, and reform between the Housing Bubble Peak, the GFC, and the Post-GFC Expansion. Any static classification scheme, GICS or network-derived, will periodically misrepresent the true correlation structure.

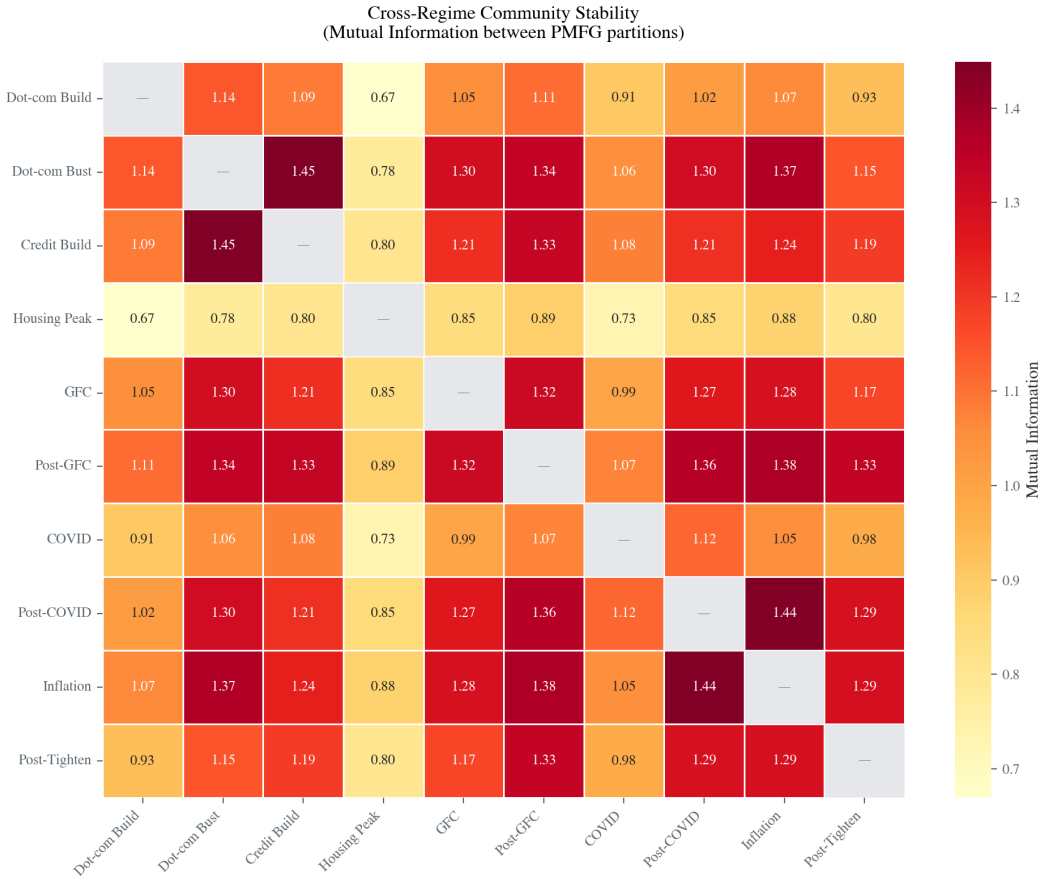


FIG. 3. Pairwise MI between PMFG partition structures across regimes. Off-diagonal values average 1.12, indicating substantial reorganisation of partition structure between any two regimes, including adjacent ones (e.g., GFC to Post-GFC: 1.17).

D. Crisis Dynamics: Within- vs Between-Sector Correlation

A direct measure of the diversification value of GICS sectors is the gap between within-sector and between-sector mean correlation. If the gap is large, stocks within a sector move together more than stocks across sectors, so sector-based diversification reduces portfolio

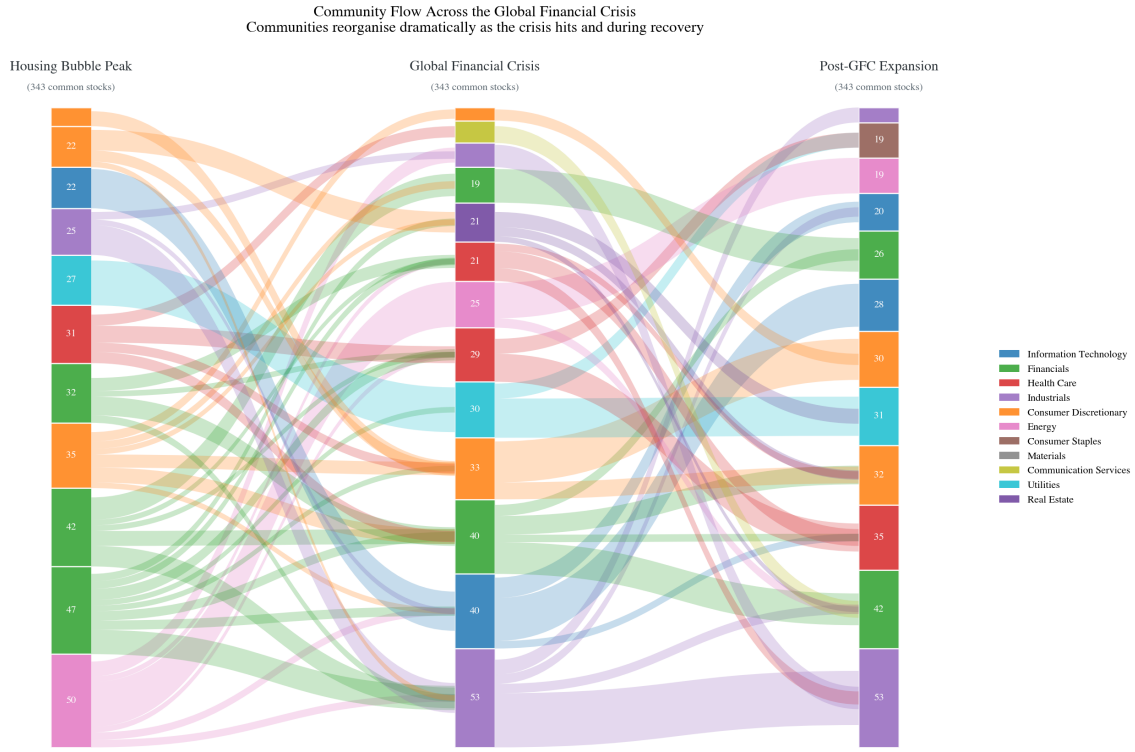


FIG. 4. Partition membership flow across the Housing Bubble Peak, GFC, and Post-GFC Expansion. Each vertical column shows partition structure for one regime; bands indicate stocks moving between partitions. The dramatic reorganisation visible here is what drives the low cross-regime MI of 1.12.

variance. If the gap closes, sector-based diversification provides no benefit.

The mean intra-sector minus inter-sector correlation gap during crisis regimes is 0.118, compared to 0.115 during expansion regimes. The crisis vs expansion gap is almost identical in magnitude. However, both absolute correlations are substantially higher during crises (mean within-sector during GFC: 0.49; during COVID: 0.55). The practical consequence is that diversification benefit from GICS sectors persists across regimes but diminishes in relative terms during crises because the correlation floor rises sharply, compressing the economically meaningful spread.

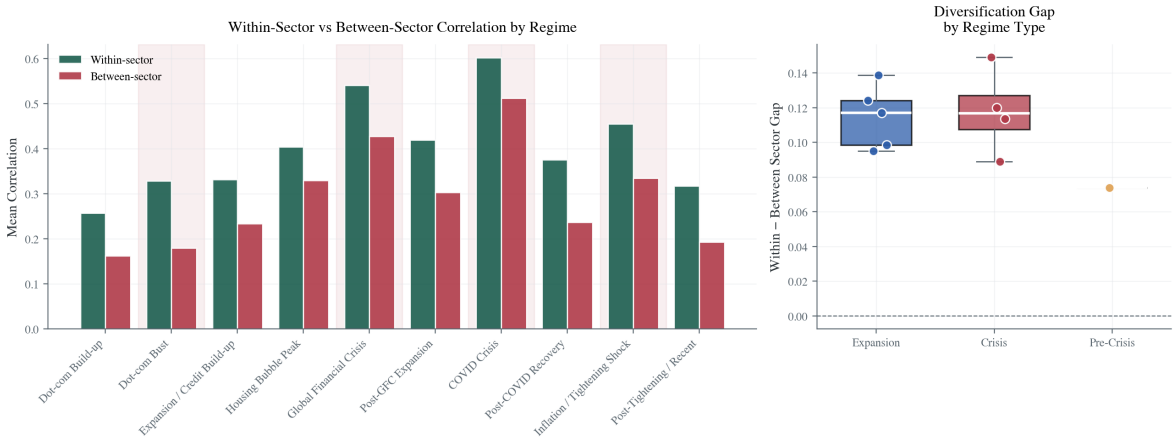


FIG. 5. Within-sector versus between-sector correlation by regime (left). The right panel shows the diversification gap (within − between) by regime type: the gap is similar in absolute terms across regimes, but both within- and between-sector correlations rise sharply in crises, eroding the relative diversification benefit.

E. Cross-Sector Co-Clustering

The sector co-clustering matrix, aggregated across all ten regimes, shows which GICS sector pairs frequently appear in the same network partition. The strongest co-clustering is Industrials–Materials (0.589), reflecting supply chain dependencies: industrial companies depend on Materials inputs (steel, chemicals, packaging), producing correlated demand cycles. Communication Services–Financials (0.419) is the second-strongest pair, reflecting the increasing prominence of advertising revenue and payment processing in both sectors. Consumer Discretionary–Industrials (0.376), Communication Services–Industrials (0.370), and Communication Services–Information Technology (0.368) follow.

Clear patterns emerge from the TRBC Activity composition of cross-sector partition members. Several "Industrials" stocks that cluster with Materials are commodity-related transportation and logistics firms whose revenues are directly sensitive to raw material prices; their TRBC Activities include "Railway Transport" and "Industrial Machinery Equipment" with exposure to mining and construction. Communication Services stocks that cluster with Information Technology include platform businesses whose TRBC Activity is "Online Services" or "Social Media Networking", stocks whose economic drivers (network effects, advertising revenue, user growth) align more closely with the growth-oriented technology

universe than with traditional media. These patterns suggest that the cross-sector clustering reflects shared economic exposures rather than idiosyncratic correlations.

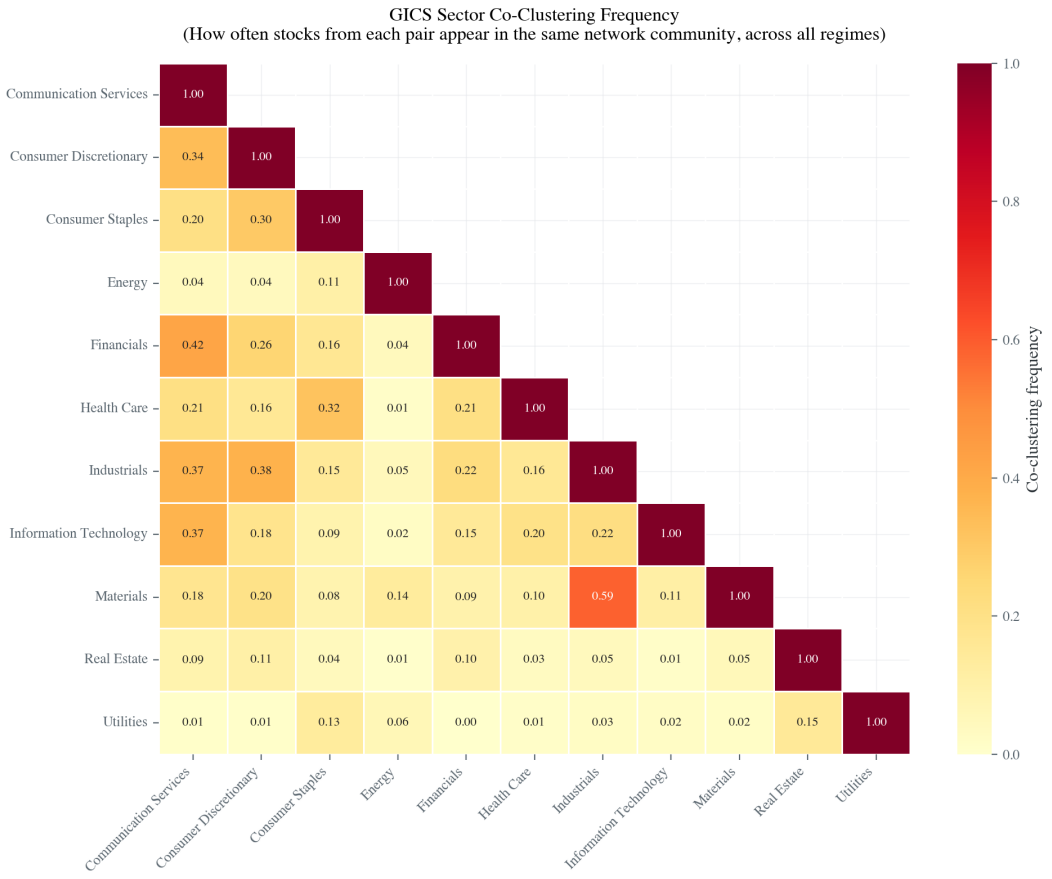


FIG. 6. GICS sector co-clustering frequency, normalised, aggregated across all ten regimes. Industrials–Materials shows the strongest cross-sector co-clustering (0.59), reflecting supply chain dependencies. Communication Services–Financials (0.42) and Communication Services–Industrials (0.37) follow.

F. Bridge Stocks and Shock Transmission

Betweenness centrality in the PMFG identifies "bridge" stocks: stocks that connect multiple partitions and along whose edges a disproportionate share of shortest paths travel. Averaged across regimes, the top-15 bridge stocks are predominantly Financials (7 of 15) and diversified Industrials (5 of 15). Within Financials, asset managers and multi-line insurers feature prominently (T. Rowe Price, Ameriprise, Prudential, Lincoln National); within

Industrials, diversified conglomerates (Honeywell, Illinois Tool Works, Emerson Electric) are the most central. This mirrors the group project's finding, at the sector ETF level, that the Industrial sector had the highest sector-to-sector betweenness centrality: the stock-level analysis identifies the same phenomenon but localises it to specific bridge companies rather than an entire sector. Figure 7 shows the PMFG for the Post-GFC Expansion regime, with the most central stocks visibly bridging partitions.

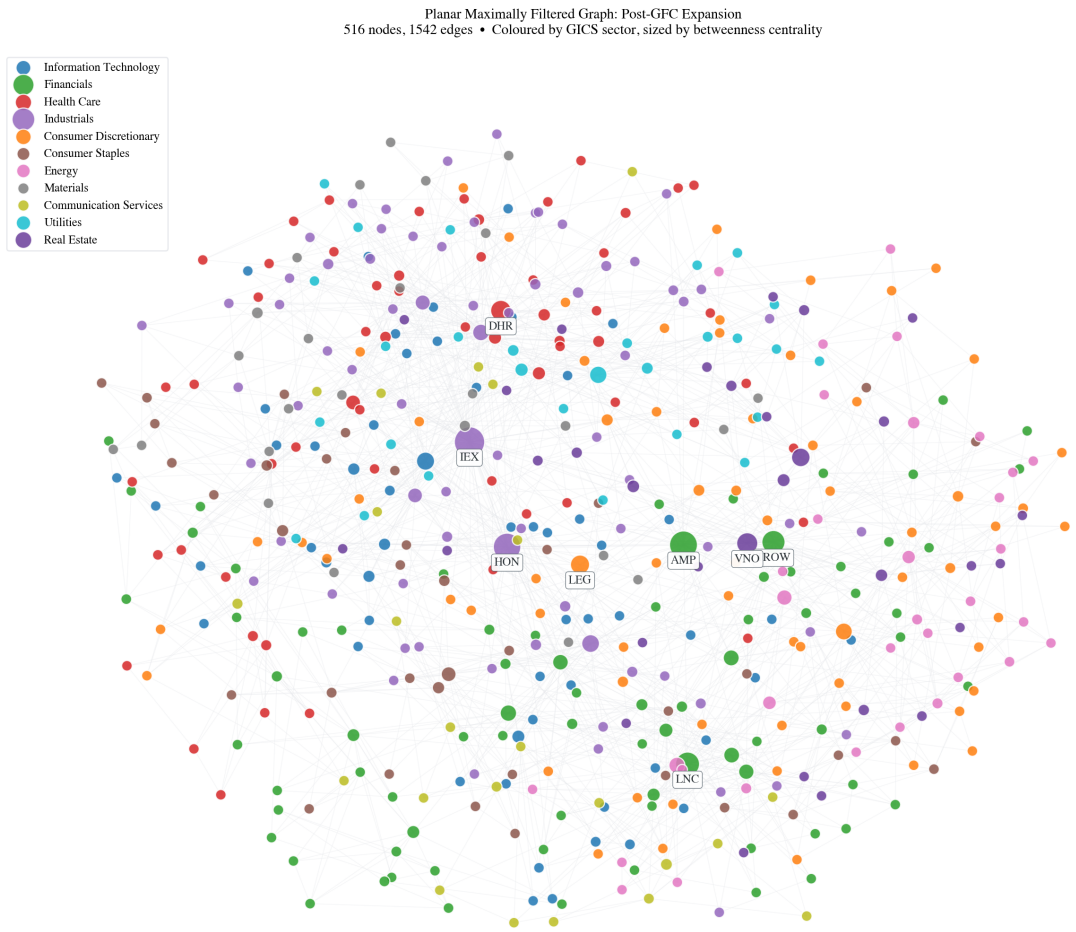


FIG. 7. PMFG network for the Post-GFC Expansion regime. 516 nodes (stocks) and 1,542 edges, coloured by GICS sector and sized by betweenness centrality. The eight stocks with highest betweenness centrality are labelled, dominated by diversified Financials (AMP, TROW, LNC) and Industrials (HON, IEX).

G. Robustness

The finding is robust to several alternative specifications. First, PMFG modularity (mean = 0.82) far exceeds that of Erdős–Rényi random graphs matched on node count and edge density (mean modularity approximately 0.45, z-scores exceeding 10 standard deviations in all regimes), confirming that the detected partition structure is non-trivial. Second, constructing the PMFG using Spearman rank correlation rather than Pearson correlation on standardised residuals yields partitions with MI $\sim$ 1.15 relative to the Pearson-based partitions, confirming the findings are not specific to linear correlation. Third, varying the Louvain resolution parameter over $\{0.5, 0.7, \ldots, 2.0\}$ changes the number of detected partitions from 5 to 22 but MI vs GICS remains within $[0.85, 1.45]$, and the per-regime ordering is preserved. Finally, repeating the analysis with the Adjusted Rand Index [10] in place of MI preserves the regime ranking (Spearman rank correlation > 0.95), indicating that the finding is not an artefact of metric choice.

H. Lead-Lag Relationships and Leadership

We apply our lead-lag framework to three key regimes with heterogeneous dynamics: the Global Financial Crisis (GFC), the COVID Crisis, and the Inflation/Tightening Shock. Each regime's network partition is converted into community return series, and the Granger causality test is applied to all ordered community pairs at each pair's optimal lag.

The Global Financial Crisis produces 29 statistically significant lead-lag pairs (out of $13 \times 12 = 156$ ordered pairs), substantially above the $\sim$ 8 expected under the null at $\alpha = 0.05$ (Bonferroni-uncorrected). The top leading community is dominated by Health Care (C9, 41 members, net leadership = +8), followed by Energy (C1, 46 members, net = +7) and Industrials (C12, 24 members, net = +3). The trailing communities are primarily Financial-dominant, which aligns with a sensible economic interpretation: during the GFC, defensive sectors (Health Care) and cyclical indicators (Energy prices, Industrial production) contained information that anticipated Financial sector movements by 5–15 trading days.

The COVID Crisis exhibits fewer significant relationships (13 pairs) consistent with its highly simultaneous nature: all sectors crashed together within weeks. The top leader is a Consumer Discretionary community (net = +3), likely reflecting early movements in stocks

directly exposed to lockdowns (travel, leisure, retail) that anticipated the broader market reaction.

The Inflation / Tightening Shock exhibits 18 significant pairs with leaders dominated by interest-rate-sensitive sectors: Real Estate (C10, 34 members, net = +4) and Utilities (C11, 32 members, net = +4). This is consistent with these sectors' direct sensitivity to the Federal Reserve's rate hike cycle, reacting before cyclicals and technology.

These findings localise and enrich the group project's sector-ETF-level lead-lag analysis. The group found the Financial sector as a crisis-period leader at the ETF level; our stock-level community analysis reveals a more nuanced picture in which different sub-sector dynamics dominate different crisis types: defensive/commodity communities lead the 2008 credit crisis, consumer communities lead the 2020 pandemic, and rate-sensitive communities lead the 2022 tightening cycle.

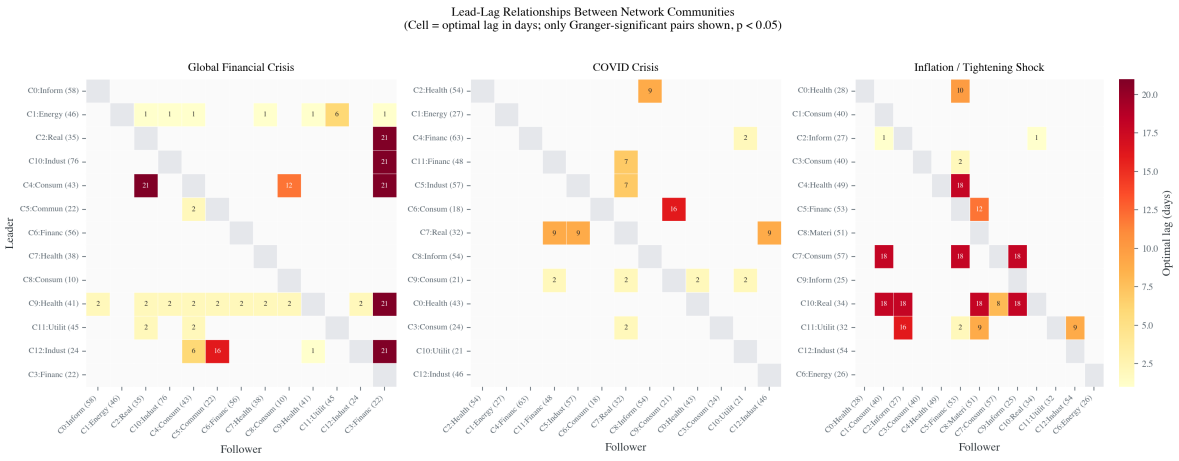


FIG. 8. Lag matrix heatmaps for the three focus regimes. Each cell shows the number of days by which the row community leads the column community, retained only when the Granger causality test rejects H_0 at $p < 0.05$. Empty cells indicate no statistically significant lead-lag relationship. The GFC exhibits the largest number of significant relationships (29), reflecting richer temporal dynamics during the slow-developing credit crisis; the COVID Crisis shows the fewest (13), consistent with its simultaneous nature.

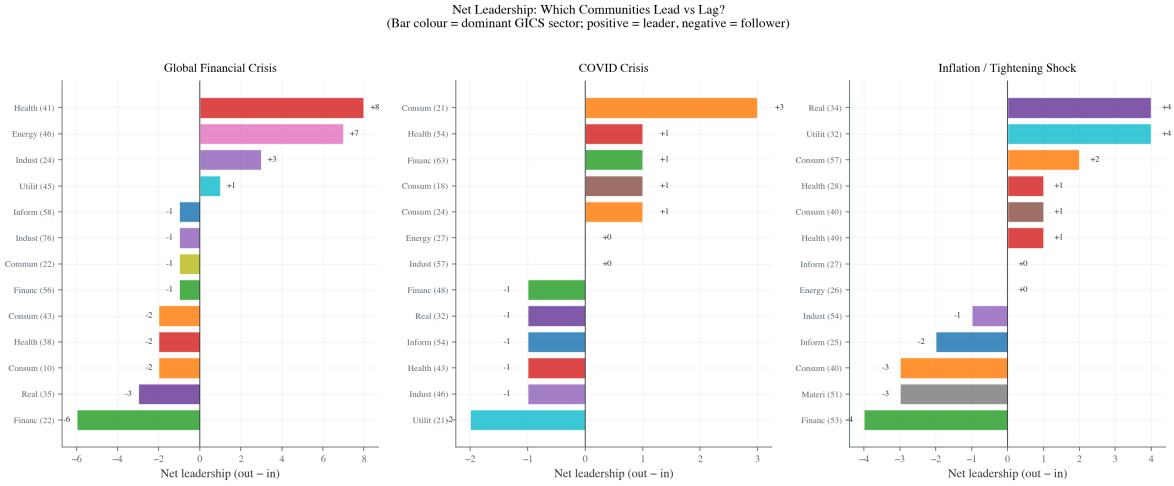


FIG. 9. Net leadership rankings per community within each regime. Bar colour indicates the dominant GICS sector of the community. Positive values identify communities with predictive information about other communities' future returns (leaders); negative values identify communities that lag behind others. The top leaders differ by regime: Health Care during the GFC, Consumer Discretionary during COVID, Real Estate during the Inflation/Tightening Shock.

V. DISCUSSION

For portfolio management, these findings have three implications. First, GICS-based diversification is not useless during crises (the within-sector premium persists) but its value is diminished at the margin because absolute correlation rises. Second, the fact that partitions reorganise across regimes (mean cross-regime MI = 1.12) means that a portfolio constructed to span distinct partitions in one regime may concentrate in a single partition during the next. Third, the identification of bridge stocks (diversified Financials and Industrials that connect multiple partitions) suggests candidate signals for contagion monitoring: an unusual rise in a bridge stock's betweenness centrality may presage a regime transition.

The results also speak to the evolving relevance of GICS. The Post-GFC Expansion and Inflation/Tightening regimes achieve the highest MI values, suggesting that during periods of relatively stable macroeconomic conditions, GICS captures correlation structure well, whereas the Housing Bubble Peak and COVID Crisis, both periods of rapid structural change, yield lower MI values (0.80 and 1.09). This is consistent with the hypothesis that the GICS taxonomy lags real-time changes in business models.

Several limitations warrant acknowledgement. First, regime definitions are *ex post* based on NBER dates and market events; real-time regime detection using Hidden Markov Models or change-point detection would be more realistic for portfolio applications. Second, while the universe now includes stocks that entered or exited the index mid-regime (reducing survivorship bias), 8 stocks required manual GICS classification due to ambiguous TRBC data; any residual misclassification may marginally bias the cohesion analysis. Third, the analysis is based on daily returns; intraday or tick-level data could reveal different partition structures driven by liquidity provision and market microstructure. Fourth, linear correlation is used throughout; non-linear dependencies (tail dependencies, asymmetric correlation) are not captured and could be relevant particularly during crises.

The lead-lag analysis adds a temporal dimension to the community detection results. The identity of leading communities changes substantially across regimes: defensive and commodity-linked communities lead the 2008 financial crisis, consumer communities lead the 2020 pandemic, and rate-sensitive communities lead the 2022 tightening shock. This regime-dependent leadership structure provides a second axis along which the static GICS classification is incomplete: even if one were to accept GICS as a fair-weather grouping, the identity of the most informative sub-groups for predicting other sectors' returns changes fundamentally between crisis types. This has direct risk-management implications: early-warning systems calibrated on one type of crisis will not transfer to another.

VI. CONCLUSION AND FUTURE WORK

This study finds that PMFG-based stock-level partitions align moderately but imperfectly with GICS sector classifications (mean MI = 1.20), with all cross-regime alignments statistically significant at $p < 0.0001$ under bootstrap resampling. Partition structures are highly unstable across regimes (mean cross-regime MI = 1.12), and within-sector correlation premium persists across regimes but both absolute correlation levels rise during crises. A Granger causality analysis at the community level further demonstrates that the identity of informationally-leading communities is itself regime-dependent: defensive and commodity-linked communities led during the 2008 GFC, consumer communities during the 2020 COVID Crisis, and rate-sensitive communities (Real Estate, Utilities) during the 2022 Inflation / Tightening Shock.

Several extensions are natural. First, the lead-lag analysis could be extended using information-theoretic measures that capture non-linear dependencies, such as transfer entropy [11], which would complement the linear Granger causality results and potentially reveal non-linear cross-community relationships invisible to the current framework. Second, detected partitions could form the basis of network-based portfolio construction, with out-of-sample backtesting against GICS-weighted benchmarks. Third, regime-switching algorithms could replace the *ex post* regime definitions, enabling real-time partition re-estimation and early warning of regime transitions.

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